Techfest 2026-27
RoboReach

Objective:

Participants must construct a manually controlled, **wireless** bot with a **mounted robotic arm** capable of completing various tasks designed to test the bot's integrity and stability, as well as the robotic arm's accuracy and control.

Bot Specifications:

- The bot must be able to fit in a box of dimensions 300 mm X 200 mm X 300 mm (l x b x h). After the run begins, the robot may extend beyond these dimensions.
- The bot's robotic arm must be able to grasp objects whose center is at a height of 450 mm (45 cm) from the ground while the bot remains on the ground. The bot's ground clearance (distance between the ground and the lowermost point of the bot) must be at least 50 mm (5 cm).
- Objects must be grasped using a mechanical gripper. Adhesives, magnets, hooks, suction cups, or other attachment mechanisms are prohibited.
- The bot must move using wheels that remain in continuous contact with the ground.
- The robotic arm must be mounted on the bot. Separate bot and robotic arms are not allowed.
- The bot shall not autonomously navigate, locate objects, or perform automatic manipulation. All drive and arm motions must result directly from operator commands. Only one person is allowed to control the bot at any point in time.
- Permitted control modes: Wi-Fi (802.11), Bluetooth (Classic or BLE), or a dedicated 2.4 GHz RF remote-control system (FHSS-based RC transmitters)
- Each robot must implement an authentication mechanism that restricts control to the team's designated controller. The robot shall reject or ignore unauthorized connection or control attempts.
- Use of mobile phones as joysticks is allowed. No internet or cellular connection is allowed during the run.
- The joysticks' dimensions are not included in the bot's dimensions.
- If the bot gets disconnected from the joystick at any time, its motors must stop, arm motion must cease, the gripper must remain at its previous position, and the bot must remain completely disabled until a new valid command is received from the joystick.

- Participants are not supposed to use any ready-made Lego components. However, the participants are allowed to use ready-made gear assemblies.
- Failing to meet any of the above specifications will lead to immediate disqualification.

Power Supply:

- The participants should use an onboard **electric supply,** i.e., the power source should be on the bot itself.
- In the case of an electric power supply, the voltage between any two points should be less than or equal to 24V DC at all times during the run.
- The battery shall be securely mounted such that it cannot become detached, even if the robot tips over or experiences sudden impacts.
- There must be a **kill-switch** on the bot that shuts it down completely, for safety purposes.
- AC power is strictly prohibited. External mains connections are incompatible with portable robots and pose safety risks during manual handling.

Arena Design & Specification:

The arena's outer dimensions are 3x3m. The course is designed to test the limits of bot stability and maneuverability and the robotic arm's precision and control.

Obstacles and Zones

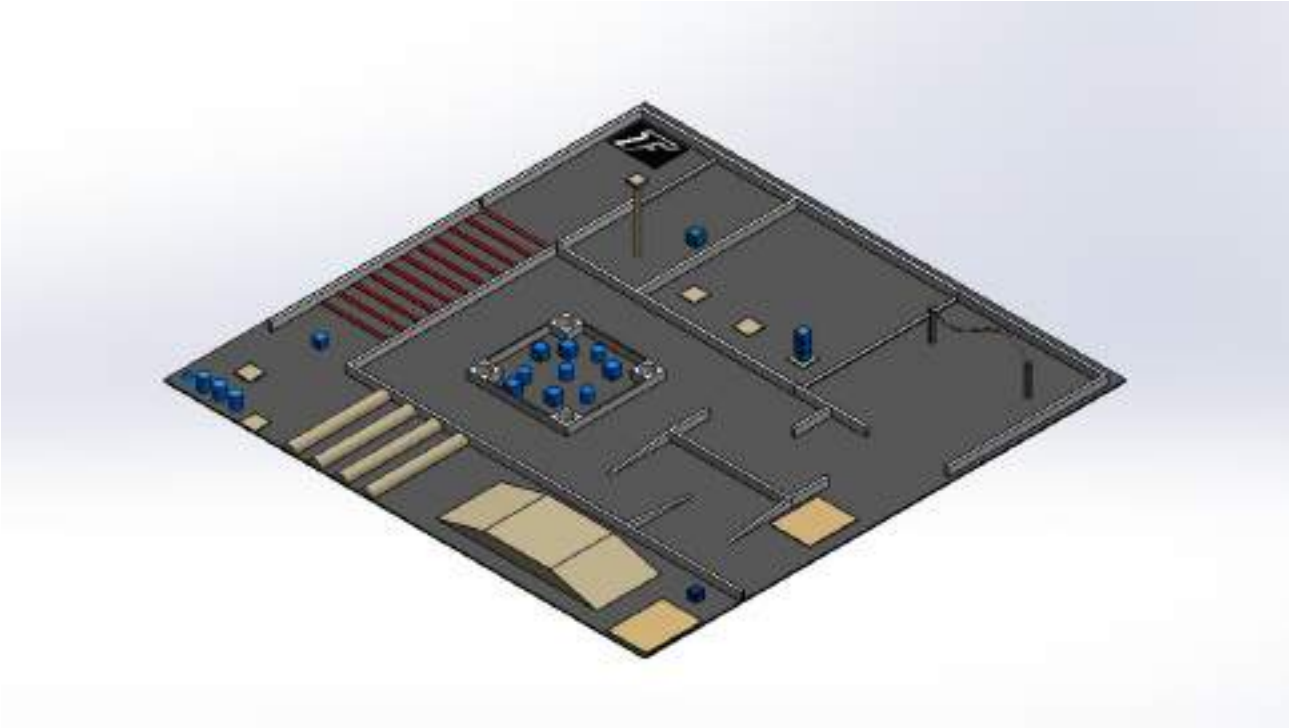
The arena is divided into distinct zones and obstacles, which the bots must navigate through.

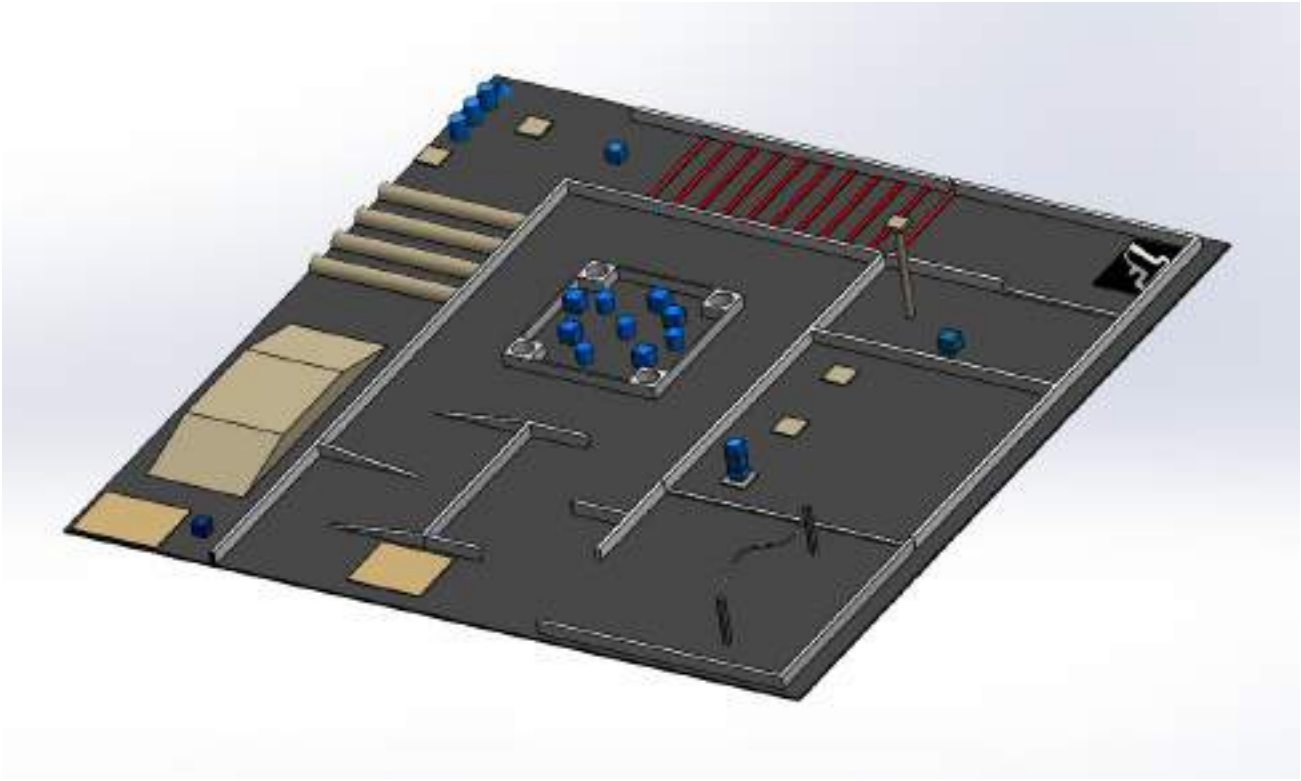
1. **The Pickup:** Pick up a cube of size 50mm x 50 mm x 50 mm from the designated place, lifting it up from the ground clearly and holding it with the robotic arm and gripper.
2. **The Ascent:** Holding the cube throughout solely with the robotic arm and gripper, ascend a ramp with a 15-degree incline.
3. **The Bridge**: Holding the cube throughout solely with the robotic arm and gripper, cross the bridge of width 400 mm.
4. **The descent breakdown**: Holding the cube throughout solely with the robotic arm and gripper, descend the bridge and immediately pass over speed breakers of peak height 40 mm.
5. **Touchdown:** Put the cube down in the designated box of size 80 mm x 80 mm.
6. **Stacker:** Pick up 3 cylinders of diameter 60 mm and height 80 mm and a cone of radius 30 mm and height 100 mm and stack them over each other with the cone at the top in the designated area marked on the arena of size 100 mm x 100

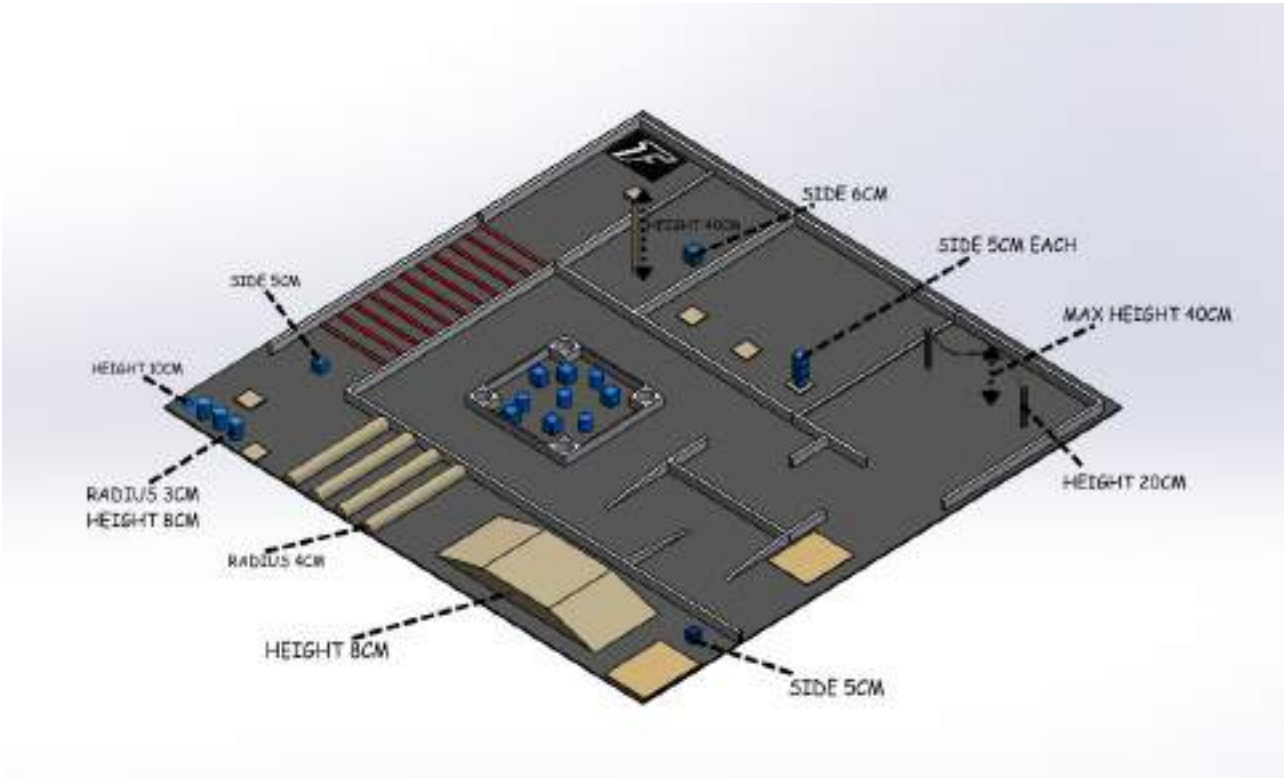
mm. This task will be considered completed when the constructed tower stands for at least 5 s after placing the last cone at the top.

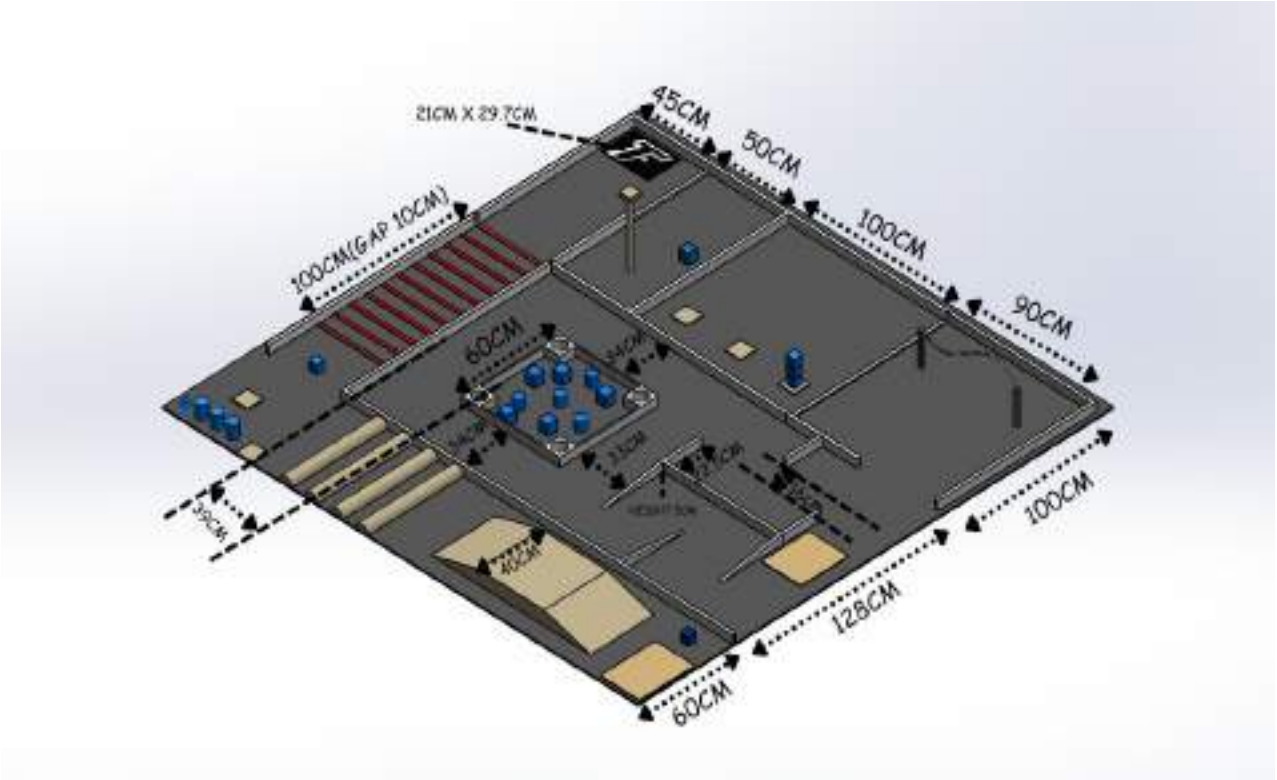
7. **Longshot:** Reach the designated area and pick up the cube of size 50 mm x 50 mm x 50 mm. Go to the throw line marked on the arena and throw the cube as far as you can. The cube could be thrown in any possible way, but the bot must not cross the throw line before the cube leaves the arm.
8. **Robonardo DaVinci:** Pick up the marker placed in the designated area and move to the paper pinned to the ground as shown in the arena map below and trace the Techfest logo inside the outline printed on the paper.
9. **Downy:** Pick up a cube of side 60 mm from the ground and place it at a height of 400 mm
10. **Hanoi'd:** Pick up (lifting the cubes clearly from the ground) and place the cubes in the designated areas of size 100 mm x 100 mm to solve the Tower of Hanoi. The blocks are numbered and are of size 50 mm x 50 mm x 50 mm. You have to move all three blocks to any of the other 2 marked squares on the ground without ever placing a block numbered higher over a lower-numbered block (like the block numbered 2 over the block numbered 1). No cube should ever be placed anywhere other than the 3 squares marked on the arena. See the link for more information.
11. **Buzzed:** Grasp a loop with a handle using solely the robotic arm and the aim is to not touch the loop to the buzz wire and move the loop to the end indicated in the arena. The height of the wire will be at most 400 mm from the ground. The buzzer will sound if the loop touches the wire at any given point like the popular Buzz wire game. This task tests the controller's control and the arm and bot's coordination.
12. **The diagonals:** Series of obstacles which limit width of track to 250 mm, testing turning of the bot in limited space.
13. **Escape Room:** Using the bot's arm, find 4 unknown objects from a box of size 600 mm x 600 mm which has many different objects. These designated objects must be identified, picked up, and dropped in the cutout of those shapes in the four corners of the box. Teams are not allowed to force the object through the hole. This task tests the arm's control, gripping ability and object manipulation. Tests the arm's precision and control.
14. **The Slalom Ridge:** A series of overlapping triangular barriers arranged in a zig-zag pattern in the path to be crossed while holding the object solely using the robotic arm, demanding sharp maneuvers and continuous dynamic stabilization.
15. **The End:** The timer will be stopped when the complete bot arrives at the designated end point of size 350 mm x 250 mm marked on the arena.

Some Images of the Arena: (*Exact Dimensions may differ by about 10%*)
Drive link of the images:
https://drive.google.com/drive/folders/1DE79dfjbRwgNwp1mGpa2VyWbZnHrOAK0?usp=sharing






SIDE 6CM
SIDE 5CM EACH
MAX HEIGHT 40CM
SIDE 5CM
HEIGHT 10CM
RADIUS 3CM
HEIGHT 8CM
RADIUS 4CM
HEIGHT 8CM
SIDE 5CM
HEIGHT 20CM



Format of Competition:

Gameplay:

- Total Time Limit: **7 minutes (420 seconds)** per run.
- Maximum Runs: Each team gets 1 run to maximize their score.
- Skipping Obstacles: Teams may skip any obstacle, but no points are awarded for skipped tasks. Skipping saves time, potentially earning a time bonus. Only one team member is allowed to keep the bot before the next obstacle under the supervision of one of the organizers.
- Time Bonus: For completion under 5 minutes (300 seconds):
    - **Bonus Points = 300 - (Total Time in seconds) (applies only if time < 300 seconds).**
    - **The maximum bonus points that can be achieved = 100.**
- Scoring Formula: Total Score = **A + T - P**
    - **A** = Total Score stored in task
    - **T** = Time Bonus Points
    - **P** = Penalties

- Task Failure: If a robot fails any task, no points are awarded for that task, but the team may continue. A task is considered failed if its completion criteria are not satisfied before the team leaves the task or the time ends.
- All objects that are to be grabbed will be made of thermocol or equivalent material.
- If the wireless bot gets disconnected from the joystick at any point during the run, one team member is allowed to handle the bot after communicating with the organizers. No repositioning, modifications, repairs, etc are allowed. There will be no penalty, but the clock will keep on running.

Obstacle Sequence and Point Allocation

Task	**Description**	**Points**
The Pickup	Pickup a cube of size 50mm x 50 mm x 50 mm from the designated place and hold it	10
The Ascent	Ascend a ramp with a 15 degree incline holding a cube	30
The Bridge	Cross a bridge of width 400 mm holding a cube	10
The descent breakdown	Descend a bridge and immediately pass over speed breakers of peak height 40 mm holding a cube	60
Touchdown	Keep the cube in the designated spot	10
Stacker	Pick up cylinders and a cone and stack them over each other.	90(20 for each cylinder and 30 for cone)
Longshot	Throw a cube of side 50 mm as far as possible.	Lines at intervals of 10 cm will be present, with 10 points for crossing each line. (100

		points max)
Robonardo DaVinci	Write TF on a piece of paper holding the marker in the robotic arm.	130
Downy	Pick up a cube placed at a height and keep it in the designated area marked on the arena.	30
Hanoi'd	Solve the Tower of Hanoi by manipulating the blocks with the robotic arm	80
Buzzed	Hold a ring with the robotic arm and move it to the end without touching the wire which passes through it	150
The Diagonals	Avoid obstacles placed on the path.	20
The Escape	Find 4 objects from a box and put them through holes of their shapes into the holes in the corner of the box.	10 points for object 1, 20 for object 2, 30 for object 3, and 40 for object 4.
The Slalom Ridge	A series of overlapping triangular barriers arranged in a zig-zag pattern.	40
The End	Stop the bot in the designated area	10

MAXIMUM POSSIBLE POINTS: 870 + 100 (Time Bonus)

Penalties

- If the bot goes out of bounds of the specified track in the arena (see figures), it will incur a penalty of 10 points per violation. The bot is considered out of bounds if any wheel leaves the track.
- After picking up the cube in task 1 (The pickup), dropping it anywhere before task 5 (touchdown) will incur a penalty of 15 points per drop. Teams are allowed to try to pick the cube again after dropping it.
- If the object is not completely within the designated area at the completion of the task, a 10-point penalty shall be applied.

- In task 7 (Longshot), overstepping the throw line will incur a penalty of 10 points.
- In task 8 (Robonardo DaVinci), a penalty of 30 points is applied if any part of either letter extends outside the printed outline.
- In task 9 (Downy), dropping the cube anywhere between picking it up from the height and placing it in its designated spot would incur a penalty of 10 points per drop. However, teams are allowed to pick up the cube again.
- In task 10 (Hanoi'd), violating any of the rules of the game of Tower of Hanoi will incur a penalty of 20 points per violation.
- In task 11 (Buzzed), touching the loop to the wire will incur a penalty of 20 points per violation.
- After picking up the object in task 13, dropping it anywhere before putting it in the box will result in a penalty of 15 points per drop. Failing to drop the object in the box through the hole will incur a penalty of 20 points per failure. However, you can try to bring a different object by going back without incurring any penalty. The bot can carry only one object at any time.

General Rules:

1. In case of any disputes/discrepancies, the organizers' decision would be final and binding
2. All bots are subject to inspections before and during the run. Organizers may require demonstration of any rule mentioned in this document.
3. Any damage done to the arena would lead to immediate disqualification.
4. Projectiles, flames, explosives, compressed gas weapons, liquids, adhesives, entanglement devices or hazardous mechanisms are prohibited.
5. The timer starts on the organizer's signal and stops when the robot completes the final task or when 420 seconds have elapsed.
6. Except where explicitly permitted, team members shall not touch the robot, arena, or game objects during a run.
7. If the bot gets stuck at any point, the bot will be placed back at the starting position of the current obstacle. No penalty will be imposed for this, but the clock will keep on running.
8. In case the bot knocks over any other task's objects, the organizers will reset them and place them at the start of the obstacle.
9. Teams shall not intentionally interfere with another team's robot, controller, wireless communications, or operation. If any such violation is found by the organizers, the team will be immediately disqualified.

10. The organizers reserve the right to disallow any design, mechanism, or strategy that exploits an unintended loophole or violates the spirit of the competition. The organizers' decision shall be final.
11. All technical, procedural, and competition-related queries must be submitted directly through the Techfest Competition Portal under the Doubt section. Participants are strictly requested to regularly check the portal for answers and updates
12. Official communication for important announcements will be through the email aryan@techfest.org. Queries submitted via email will not be entertained

Tie-Breaker Rules:

1. Fastest Completion Time: If the number of tasks completed is equal, the bot with the lower completion time is ranked higher.
2. Fewer penalties: If the teams have the same completion time, the team incurring fewer penalties will be ranked higher.
3. If there is still a tie, both teams will be given the same rank.
4. High-Difficulty Task Completion: If completion times are also equal, the bot that completes more high-difficulty tasks is ranked higher. Task difficulty is determined by the points assigned to each obstacle in the **Obstacle Sequence & Point Allocation table**, a higher point value indicates a higher-difficulty task.

Team Specifications & Eligibility:

- **Eligibility:** Open to all students with a valid ID from their educational institution.
- **Team Size:** Teams may include students from different institutes. Maximum of 4 members per team.

Eligibility Criteria:

- All students with a valid identity card from their respective educational institutes are eligible to participate

Timeline:

Event	Deadline
Registration closes	As per the date of the respective city zonal
Zonals Qualifiers Round	On the date of that city zonal
Results of Zonals winners	On the day of that city zonal
Finals	15-18th December

Certificate Policy:

- The **Top 5** teams per zonal city advancing to Round 2 will receive a **Certificate of Excellence**.
- E-Certificates of participation will be given to the teams scoring more than the critical marks, which will be decided later

Prize Money:

The Prize Money will be awarded to the top 5 winners of the final round via NEFT and will be processed by May 2027.

Prize Money Submission Process

- **Submission Channel:** Complete the prize money form **directly on the website** after the final round concludes. The winners would be able to view the form through their registered email ID
- **Deadline:** The form must be submitted by **December 31, 2026**.

Details Needed for the Website Form

When filling out the form on the website, make sure you have the following bank account details ready:

1. Account Holder's Name
2. Account Number
3. Bank Name and Branch Name
4. IFSC Code